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Multiple Authority Announcement (MAA)

To access hyperlinks from this electronic solicitation – Hit CTRL and click on the link.

1.0 Overview Information

Federal Agency Name	Air Force Research Laboratory, Sensors Directorate, Multi-Domain Sensing Autonomy Division, AFRL/RYA
MAA Title	Autonomous Decisions, Algorithms, and Modeling (ADAM)
MAA Number	FA2377-25-S-1207
MAA Type	This is the Initial Announcement
Competition Type	'Other Competitive Procedure' as described in FAR 6.102(d)(2), FAR 35.016, DFARS 235.006-71, as well as 2 CFR §200.203, 10 USC §4001 and 10 USC §3458
MAA Process	The MAA process consolidates several solicitation authorities under a single announcement. As AFRL/RYA Objectives are identified, individual Solicitations will be released under this overarching Announcement. Individual Solicitations may be issued as Calls for Proposals OR Open Periods AND will utilize either a One- OR Two-Step format. See Attachment 1, ADAM MAA Industry Guide for additional details.
White Paper/Proposal Due Date	<u>NO PROPOSALS ARE REQUESTED AT THIS TIME.</u> Potential offerors are invited to monitor this announcement for the release of individual solicitations which will detail specified Air Force Objectives. Each individual solicitation will identify White Paper/Proposal due date(s) and time(s).
Anticipated Award Date	TBD per individual solicitation.
Estimated Program Cost	TBD per individual solicitation.
Anticipated Number of Awards	TBD per individual solicitation.
North American Industry Classification System (NAICS)	The NAICS Code for this acquisition will be TBD per individual solicitation. However, it is anticipated most efforts will utilize 541715 (Research and Technology in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology)) , and the small business size standard is 1,000 employees .
Federal Assistance Listing (Formally Catalog of Federal Domestic Assistance (CFDA)) Number(s)	12.800 AIR FORCE DEFENSE RESEARCH SCIENCES PROGRAM
Agency Contacts	Address general MAA questions to: AFRL.RYA.ADAMmaa@us.af.mil

2.0 Full Text Announcement

2.1 Program Description:

The AFRL/RYA ADAM MAA is intended to provide a comprehensive strategy for AFRL/RY's range of S&T, allowing for progression from basic research to technology maturation and transition.

2.2 Solicitation Authority:

Solicitations issued under this MAA will utilize any one, or a combination, of the authorities below:

- 10 U.S.C. 4001 and Federal Acquisition Regulation and its Supplements (FAR(S))
- 10 U.S.C. 4001 and 10 U.S.C. 3458 - Authority to acquire innovative commercial products and commercial services using general solicitation competitive procedures
- 10 U.S.C. 4021 – Research projects: transactions other than contracts & grants (Other Transaction for Research)
- 10 U.S.C. 4022 – Authority of the Department of Defense to carry out certain prototype projects (Other Transaction for Prototype).
- 10 U.S.C. 4023 – Procurement for Experimental Purposes
- 10 U.S.C. 4001 and 32 CFR 22.215 – Grants and Cooperative Agreements

2.3 Statement of Objective/Description of Technical Area(s):

Solicitations issued under this MAA will seek approaches covering one or more of the following Technical Areas. Efforts awarded under these Technical Areas may consist of concept development, analysis, basic research, applied research, advanced technology development, advanced component development and prototyping, experimentation and/or transition activities that will enable future efforts.

2.3.1 Technical Area 1: Multi-Domain Sense Making: Efforts in this area use available data and information to describe, explain, and predict the state of the world. This process meaningfully translates relevant data into usable information and information into knowledge. Efforts will use data, modeling & simulation, algorithms (physics-based, AI/ML, or hybrid), computational resources, and test & evaluation. Each effort works to develop and transition military relevant AI/ML technology to fuse information from any source, reasons over the environment, and enables improved, timely, and executable battlespace decision making. Efforts include sensemaking from single sensors, multiple sensors, modalities, and domains applied at-the-edge, at forward operating units, and/or in reach-back cells.

2.3.2 Technical Area 2: Sensing Autonomy: Efforts in this area focus on ISR asset management, exploitation, and control, as well as support mission success through tailored application of platforms, sensors, infrastructure, and analysis capabilities. These applications may be on premise, in cloud-based architectures, and at the edge. Efforts in this area develop and apply sensing management principles to accommodate multi-domain information sources to accelerate sensing & analysis. Efforts also enable multiple forms of autonomy in motion and at rest within and as a component of systems, processes, and artifacts within the Air Force's aggressive Digital Transformation efforts and intentions. Efforts will require research in sensor data processing and the underpinning theories. Efforts will require expertise in MS&A, along with creation, cataloging, reuse and repurposing of digital artifacts, software engineering, software development, and cyber secure software integration and testing.

2.3.3 Technical Area 3: Sensing and Effects Analysis: Efforts in this area prepare the warfighter by improving ISR products across all domains with performance understanding, mission effects analysis, weapon assessments and modeling & simulation (M&S) to exploit heterogeneous networked sensors through artificial intelligence, machine learning and sensor fusion.

2.3.4 Technical Area 4: Multi-Sensing Knowledge: Efforts in this area provide techniques for timely, high confidence behavioral and physical knowledge generation from denied and difficult targets using multiple sensors, domains, and types to include algorithm development across multiple distributed, homogeneous and heterogeneous sensors. Efforts will commonly include data association, entity detect/track/ID, information fusion, contextual reasoning, training with limited measured data, data/performance modeling, and scenario specific algorithm performance assessment. Leads the application of machine learning techniques to address technical challenges in contested environments.

2.3.5 Technical Area 5: Sensing Management: Efforts in this area develop multi-domain strategies to orchestrate closed-loop sensing that effectively manages environment understanding to effects across multiple ISR, Strike and EW/Cyber mission effects chains. These efforts include techniques to manage sensor data flow through collection, communication, and reasoning for processing and dissemination; to generate anticipatory responses; sensor resource planning, allocation, and scheduling; and control flexibility across multiple distributed sensing capabilities. Efforts will focus on technologies including sensing interface/architecture development and assessment, experimentation, sensing decision-making strategies, representation, sensing data and knowledge management, cross-mode sensor management and registration, distributed processing, and joint inference and control.

2.3.6 Technical Area 6: Efforts in this area provide techniques for behavioral and physical knowledge generation of targets from multiple sensors, domains, and sensor types. Efforts will include sensor exploitation and autonomy algorithm development, assessment, and experimentation across multiple distributed, homogeneous, and heterogeneous sensing phenomena. This research area focuses on technologies required in support of sensing autonomy technology development including ML/AI, sensing architecture development and assessment, sensing decision-making strategies, sensing performance modeling, target and sensing scenario knowledge representation, sensing data and knowledge management, distributed processing, high performance computing, and joint inference and control.

2.4 Communication Between Prospective Offerors and Government Representatives:

The communication plan with prospective offerors will be multi-pronged and tailored to specific solicitation needs. The solicitation specific communication plan may include hosting industry days, releasing Requests for Information (RFIs), conference engagements, and direct notifications for classified solicitations.

Address technical questions to the Technical POC: TBD per Solicitation.

Address contracting questions to the Contracting POC: TBD per Solicitation.

2.5 Within Scope Modifications:

Potential offerors are advised that due to the inherent uncertainty of research and development and technology transfer efforts, awards resulting from this announcement may be modified during performance to make within scope changes, to include but not limited to,

modifications which increase overall contract ceiling amount and modifications under authority of DFARS 235.006-71. Modifications will be governed by the applicable authority.

2.6 Deliverable Items: TBD per Solicitation.

2.7 Other Requirements:

- a. Program Security Classification: TBD per Solicitation. If a DD254 is applicable, offerors must verify their Cognizant Security Office information is current with Defense Counterintelligence and Security Agency (DCSA) at <http://www.dcsa.mil>.
- b. TEMPEST Requirements: TBD per Solicitation.
- c. OPSEC: General Operations Security (OPSEC) procedures, policies and awareness are required to reduce program vulnerability from successful adversary collection and exploitation of critical information. OPSEC will be applied throughout the life cycle of any effort awarded as a result of Solicitations issued under this MAA. The Critical Information & Indicators List (CIIL) and the OPSEC Plan will be provided upon request by AFRL DSO Security Manager Office. While working on the government installation, OPSEC guidance and OPSEC training will be provided by AFRL DSO Security Manager Office. This training will ensure contractors are familiar with CIIL and OPSEC Plan as it pertains to their contract. The contractor shall apply OPSEC in the management of their program IAW AFI 10-701 Operations Security and WPAFB Supplement to AFI-10-701.
- d. Export Control: Information involved in awarded efforts **may** be subject to Export Control (International Traffic in Arms Regulation (ITAR) 22 CFR 120-130, or Export Administration Regulations (EAR) 15 CFR 710-774). Individual Solicitations will specify applicable Export Control requirements. If Export Control applies, then a Certified DD Form 2345, Militarily Critical Technical Data Agreement, will be required to be submitted with proposal. Only contractors who are registered and certified with the Defense Logistics Agency and have a legitimate business purpose may participate in this solicitation. Contact the U.S./Canada Joint Certification Program Office, Defense Logistics Agency, Logistics Information Services J34, HDI Federal Center, 74 Washington Avenue N., Battle Creek, Michigan 49037-3084, (1-877-352-2255) or the Joint Certification Program Office (JCO) at JCP-ADMIN@DLA.MIL for further information on the certification process. You must submit a copy of your approved DD Form 2345, Militarily Critical Technical Data Agreement, with your proposal, if required by the solicitation.

2.8 Science and Technology (S&T) Protection:

S&T Protection requirements are governed by Air Force Research Laboratory Instruction (AFRLI) 61-113, "*Science and Technology (S&T) Protection for the Air Force Research Laboratory*". Applicable S&T Protection requirements will be identified in each Solicitation.

2.9 Other Information:

2.9.1 Government Furnished Property (GFP) Availability:

TBD per Solicitation. In accordance with FAR 45.201(b), the contractor is responsible for all costs related to making the property available for use, such as payment of all transportation, installation or rehabilitation costs.

2.9.2 Base Support/ Network Access:

TBD per Solicitation.

2.9.3 Multiple awards subject to Fair Opportunity:

TBD per Solicitation.

2.10 Contract Type/Instrument:

The Air Force may make award via any appropriate instrument allowed by the Authority utilized in the applicable Solicitation. Solicitations may contemplate the award of FAR-based contracts, grants, cooperative agreements, commercial contracts, Other Transactions for Research (OTR), Other Transaction for Prototype (OTP), or Other Transactions for Experimentation (OTE). The Air Force may also consider award of an appropriate technology transfer mechanism if applicable. It is anticipated that awards will generally be cost reimbursement contracts (Cost-Plus-Fixed-Fee, Cost, or Cost Share) or expenditure-based OTs, but this will be TBD per Solicitation. For FAR-based awards, cost reimbursement contracts require successful offerors to have an accounting system considered adequate for tracking costs applicable to the contract.

This Announcement permits the award of initial production or initial or additional items pursuant to the authority at DFARS 234.005. In addition, this Announcement provides for the award of Prototype OTs and follow-on production contracts or transactions pursuant to 10 USC 4022. Consequently, offerors and proposers are notified of the potential for follow-on production contracts or transactions without the use of competitive procedures.

2.11 Data Rights Desired:

TBD per Solicitation. See ADAM MAA Industry Guide, Chapter 3 for additional details.

3.0 Award Information**3.1 Anticipated Award Date:**

TBD per Solicitation.

3.2 Anticipated funding:

TBD per Solicitation.

Any funding profiles provided should be considered estimates only and not a contractual obligation for funding. All funding is subject to change due to Government discretion and availability. Potential offerors should be aware that due to unanticipated budget fluctuations funding in any or all areas may change with little or no notice.

4.0 Eligible Offeror:

It is anticipated that Solicitations issued under this MAA will be unrestricted. Small businesses are encouraged to propose.

4.1 Cost Sharing or Matching:

TBD per Solicitation.

4.2 Federally Funded Research and Development Centers:

The following guidance is provided for Federally Funded Research and Development Centers (FFRDCs) contemplating submitting a proposal in response to a Solicitation, as either a prime or subcontractor. FAR 35.017-1(c)(4) prohibits an FFRDC from competing with any non-FFRDC concern in response to a Federal agency request for proposal for other than the operation of an FFRDC (with exceptions stated in DFARS 235.017-1(c)(4)). There is no regulation prohibiting an FFRDC from responding to a Solicitation issued under this MAA. However, the FFRDC's sponsoring agency must first make a determination that the effort being proposed falls within the purpose, mission, general scope of effort, or special competency of the FFRDC, and that determination must be included in the FFRDC's proposal. In addition, AFRL must make a determination that the work proposed would not place the FFRDC in direct competition with domestic private industry. Only after these determinations are made, would a determination be made concerning the FFRDC's eligibility to receive an award.

4.3 Government Agencies:

If a Government agency is interested in performing work, contact the Program Manager/ Technical POC identified in the Solicitation. If those discussions result in a mutual interest to pursue your agency's participation, the effort will be pursued independent of that Solicitation.

4.4 Other:

4.4.1 Foreign Participation:

Notice to Foreign-Owned Firms: Such firms are asked to immediately notify the Contracting POC identified in the Solicitation before deciding to respond. Foreign contractors should be aware that restrictions might apply which could preclude their participation.

4.4.2 White Paper/Proposal Limits:

While not anticipated, Solicitations will determine any limits on the number of white papers/proposals an offeror may submit. If permitted by the Solicitation, and an offeror submits multiple white papers/proposals, each white paper/proposal must provide unique approaches.

4.5 Ineligibility:

Offerors may be ineligible for award if all requirements of this Announcement and applicable Solicitation are not met by any identified due date or Solicitation closing date.

5.0 White Paper / Proposal and Submission Information

See ADAM MAA Industry Guide Chapter 4.

6.0 White Paper / Proposal Review Information

See ADAM MAA Industry Guide Chapter 5.

7.0 Award Administration Information

7.1 Award Notices:

Solicitations will identify when Offerors will be notified whether their proposal is recommended for award. The notification is not to be construed to mean that any contract, agreement or assistance award is assured, as availability of funds and successful negotiations are prerequisites to any award.

7.2 Administrative and National Policy Requirements:

See Section 2.0. Additional requirements may be TBD per Solicitation.

7.3 Reporting:

Specific reporting requirements attributable to individual authorities will be identified in the Solicitation.

8.0 Other Information

8.1 Acquisition of Commercial Items:

Solicitations will determine whether or not the Air Force intends to use the policies contained in Part 12, Acquisition of Commercial Items. However, in any case, interested offerors may identify to the Contracting Officer their interest and capability to satisfy the Government's requirement with a commercial item within 15 days of Solicitation release.

8.2 Support Contractors:

Only Government employees will participate in Peer or Scientific Reviews. Offerors are advised that employees of commercial firms under contract to the Government may be used to administratively process proposals, monitor contract performance, or perform other administrative duties requiring access to other contractors' proprietary information. These support contracts include nondisclosure agreements prohibiting their contractor employees from disclosing any information submitted by other contractors or using such information for any purpose other than that for which it was furnished.

8.3 Feedback Sessions:

- a. **Contracts:** When requested, a Feedback Session will be provided with content consistent with the procedures that govern BAAs (FAR 35.016 or applicable regulation). The process will follow the time guidelines outlined in the award notice described in Paragraph 7.1.
- b. **Assistance Instruments and Other Transactions:** When requested, an informal feedback session will be provided. The process will follow the time guidelines outlined in the award notice described in Paragraph 7.1.

List of Attachments:

1. ADAM MAA Industry Guide
2. Statement of Work (SOW) Supplemental Requirements